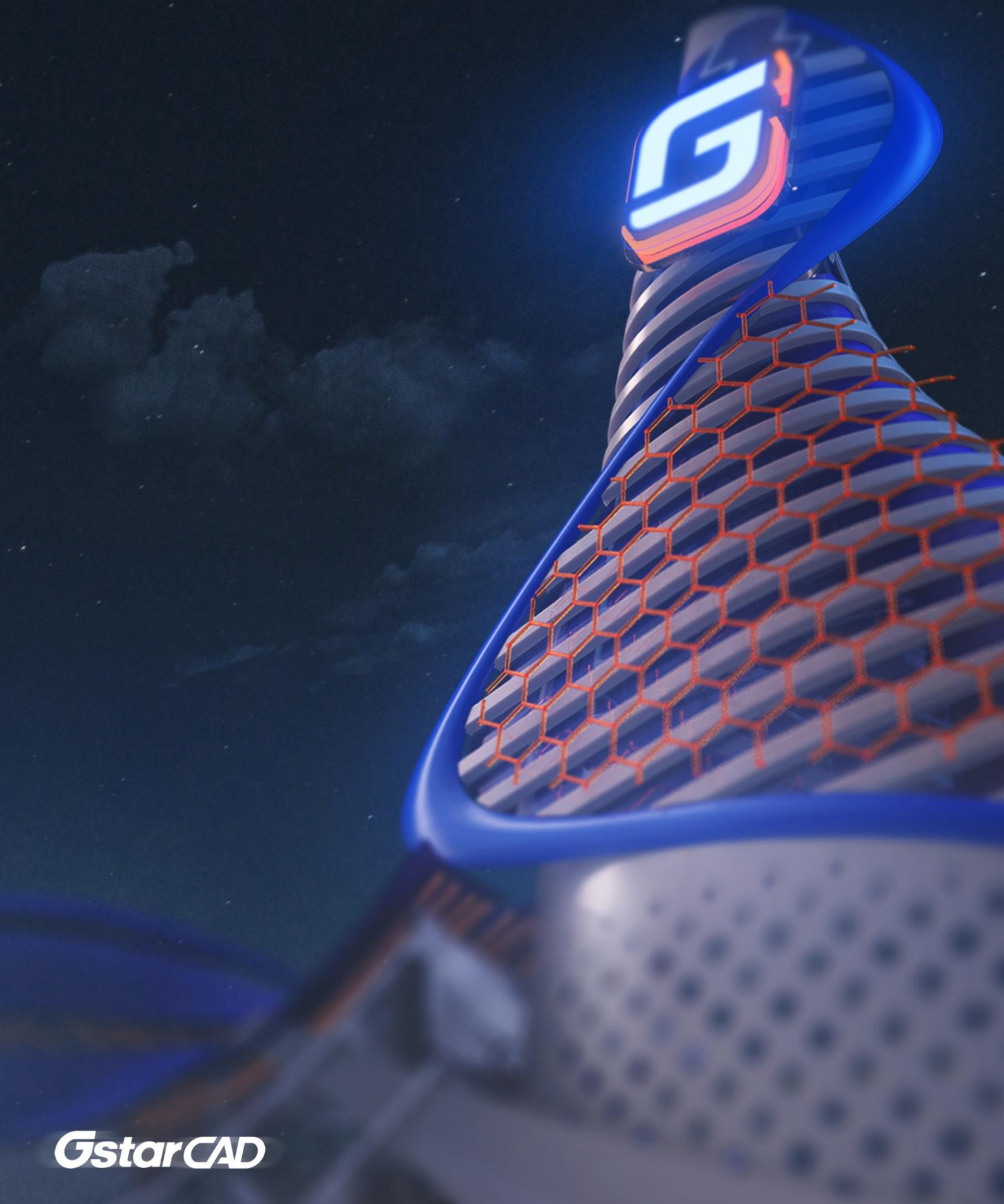
.NET Upgrade Guide

GstarCAD 2020



GstarCAD

Table of Contents

1. Brief Introduction 3
2. Development Environment 4
3. VS2017 C# .NET Program Upgrade Process 5
   3.1. Change Reference Files 5
   3.2. Compile Program 9
4. VS2017 VB.NET Upgrade Process 10
   4.1. Change Reference Files 10
   4.2. Compile Program 14
5. Copyright 15

1. Brief Introduction

This document describes how to upgrade the .Net project/application from a lower version of GstarCAD to GstarCAD 2020.

2. Development Environment

- Microsoft Visual Studio Enterprise 2017（Version 15.9.17）
- CPU：

Basic Requirement：1.6 GHz CUP，Recommend：3.0 GHz CPU and above

- RAM：

Basic Requirement：2 GB，Recommend：8 GB and above

- Operation System

Windows 10 version 1507 and advanced version：Including home version, professional version, education version and enterprise version（not support LTSC and Windows 10 S）

Windows 8.1（with updated 2919355）： Core version, professional version and enterprise version

Windows 7 SP1（with latest Windows updated）： Including home version, professional version, enterprise version and Ultimate version

- Monitor Resolution:

Traditional monitor：1028 x 800 and above CRT monitor, High resolution and 4K (3840 x 2160) monitor

- GRX SDK 2020
- GstarCAD 2020
- .NET Framework 4.8 and above

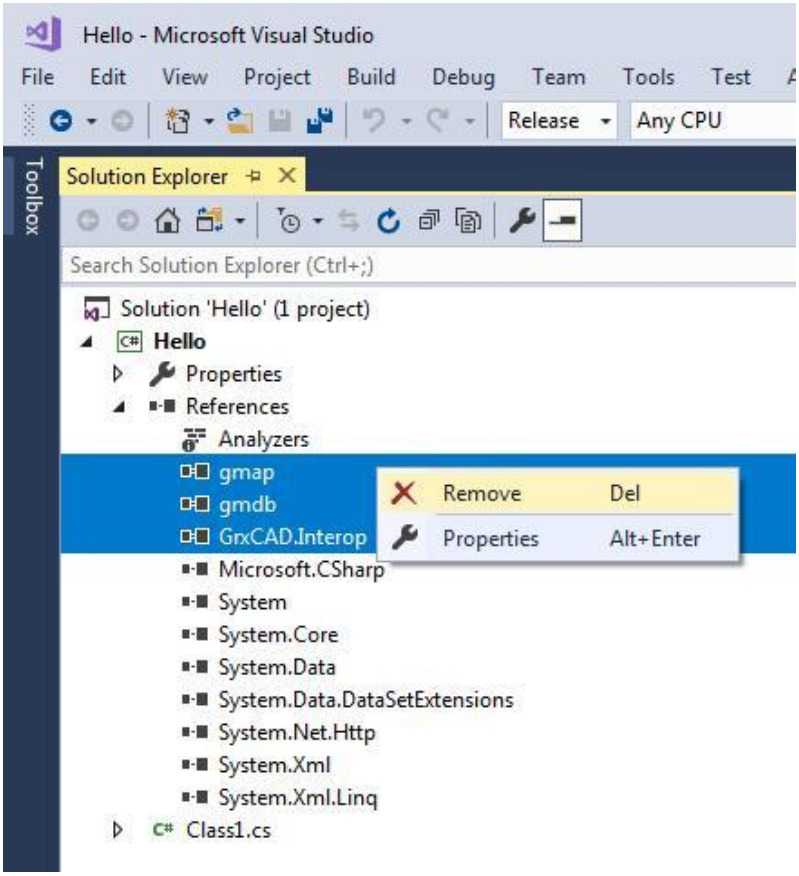
3. VS2017 C# .NET Program Upgrade Process

The following is a sample to create a "Hello" project, describe the creation process and suppose GstarCAD SDK package installed in c:\grxsdk dictionary.

3.1. Change Reference Files

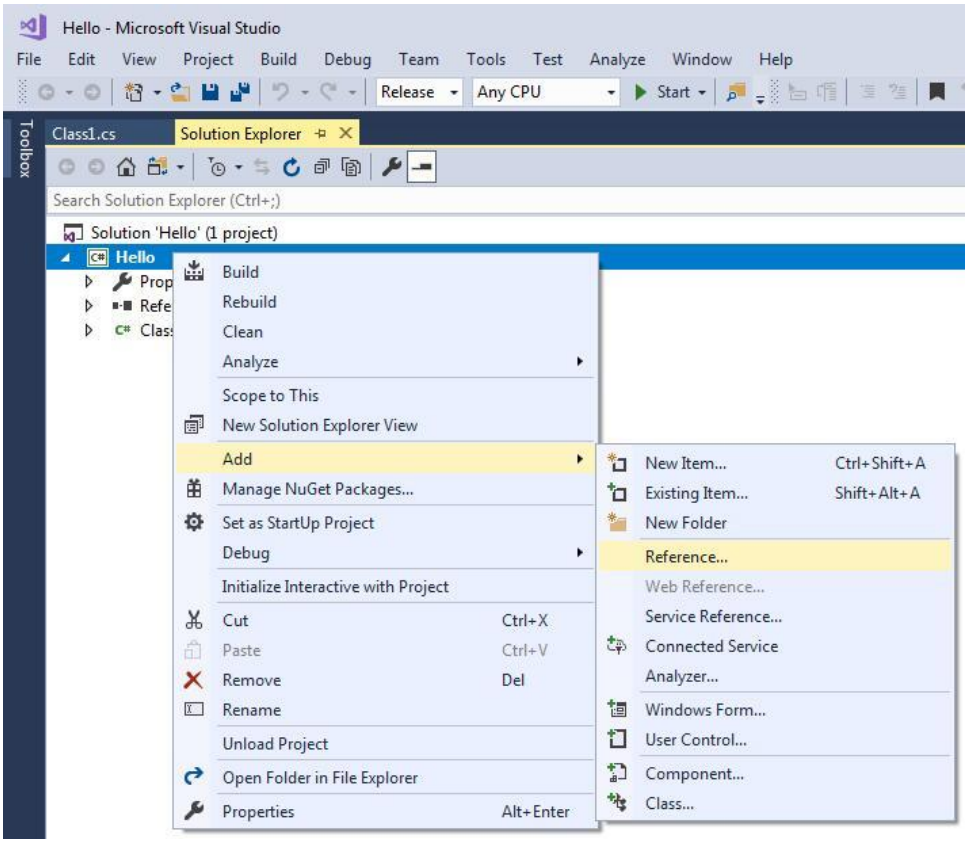
3.1.1. Remove the Reference

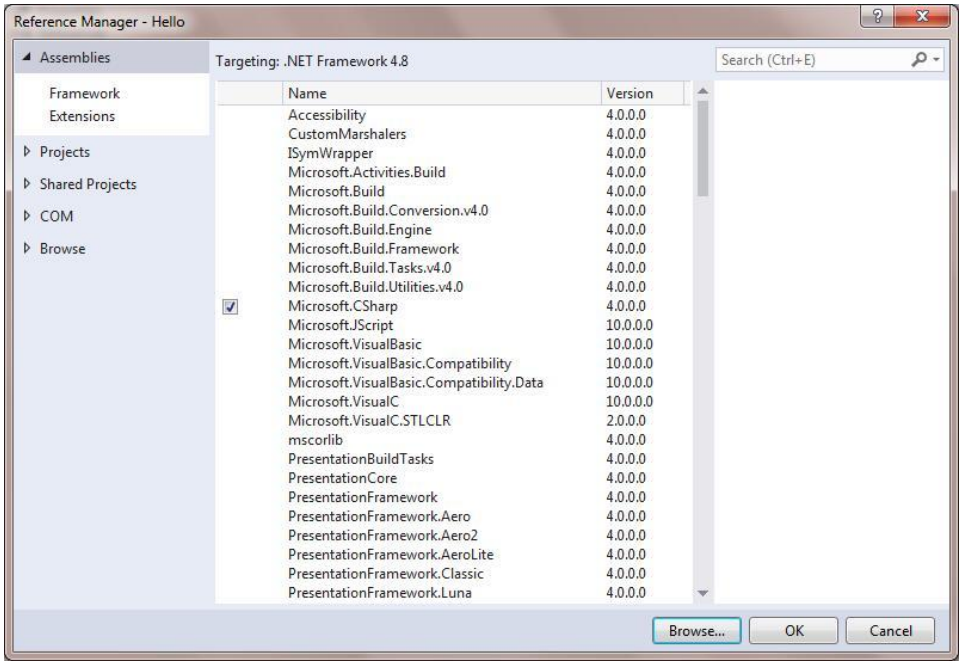
Right click on the reference files that referenced to the previous version of GstarCAD, click Remove.



3.1.2. Add Reference

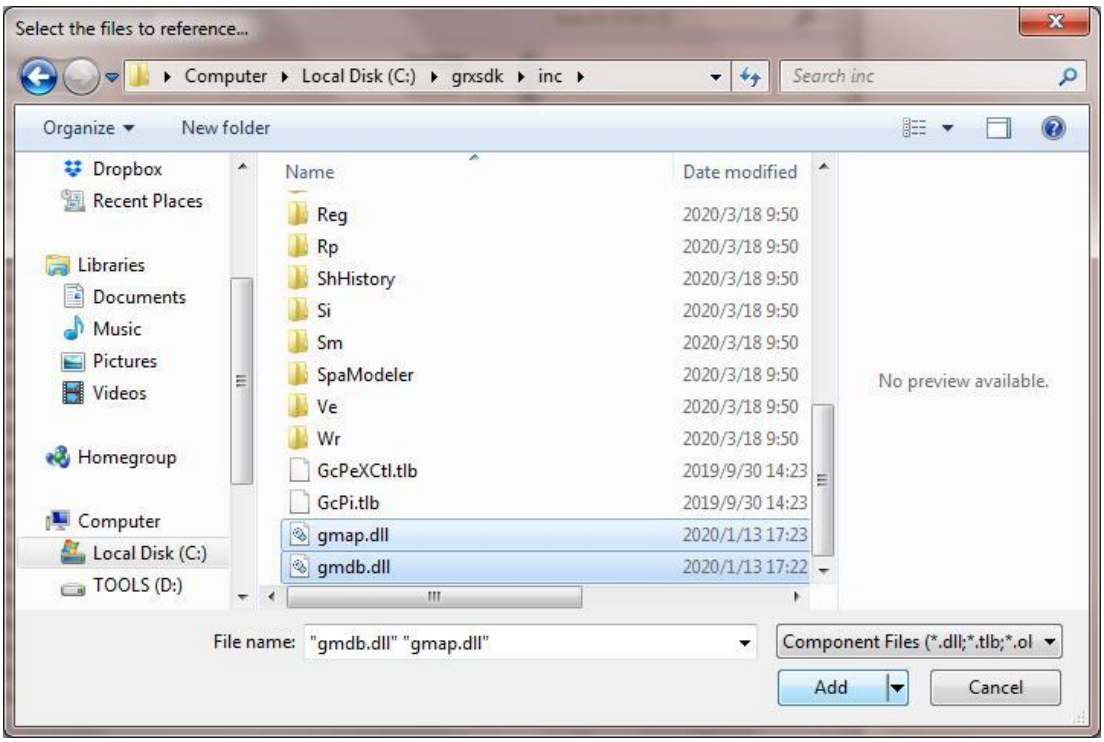
In VS 2017, find the Hello project you have been created on the solution explorer. Select the Hello project and right click to select Add Reference. After the "Reference Manager" dialog box pops out, click Browse button as shown below.

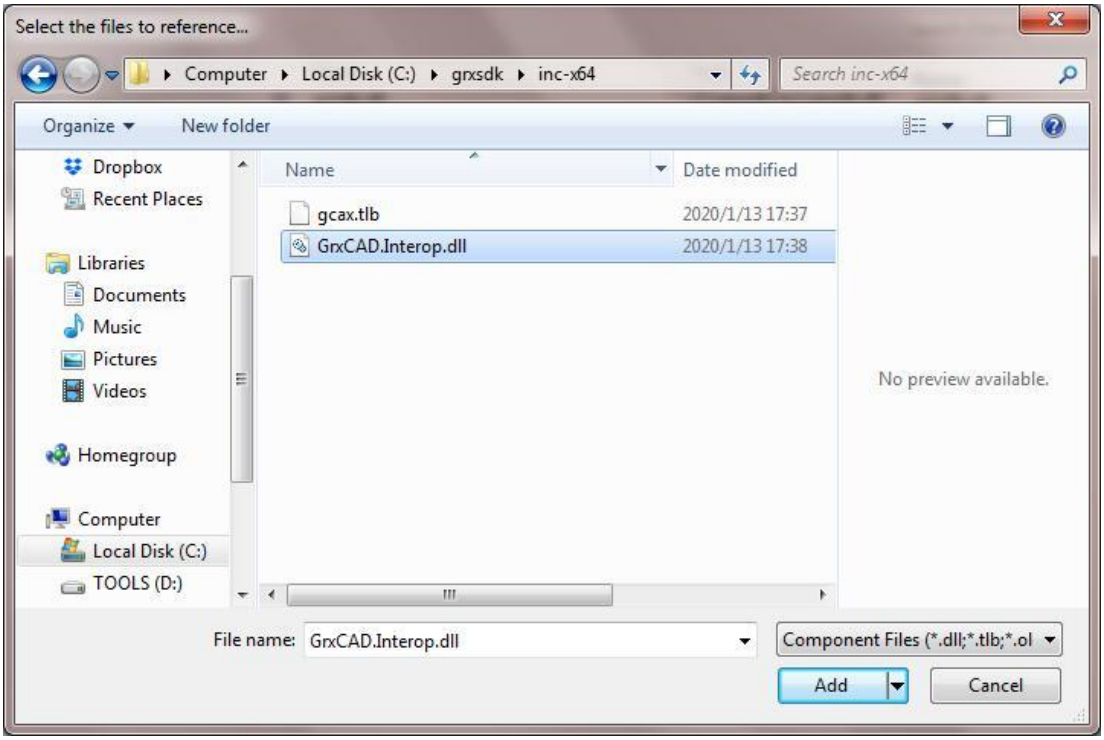




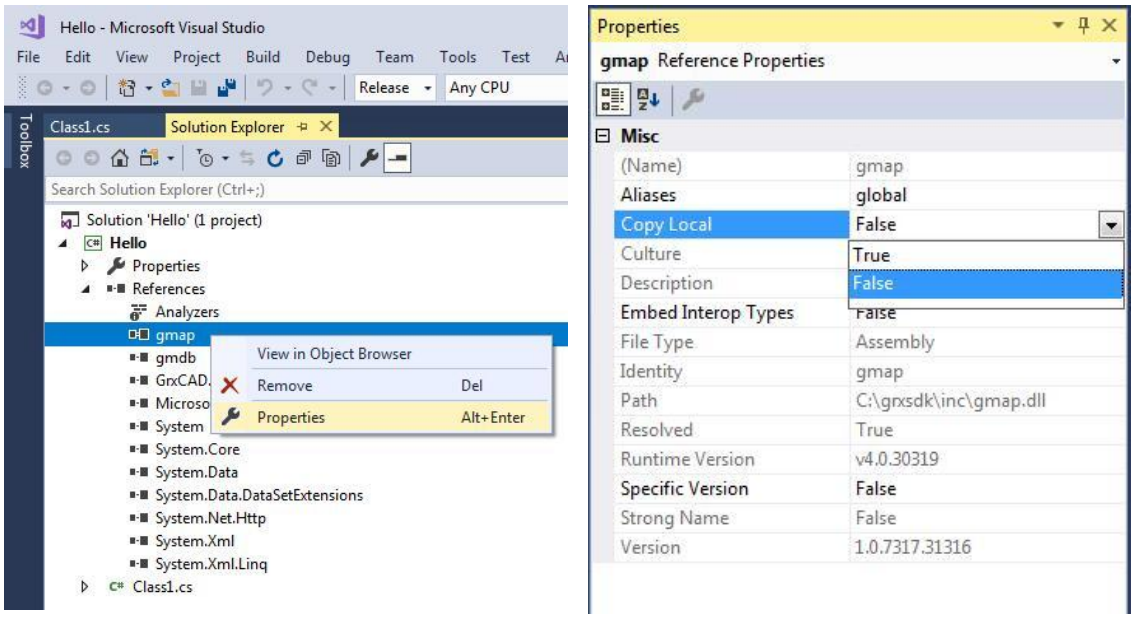
3.1.3. Choose and Add the Files

Find the GstarCADSDK installation path, for example, “C:\grxsdk\inc”. Select gmap.dll and gmdb.dll files in “c:\grxsdk\inc-x86” and add them by clicking Add button（If you use GstarCAD 64-bit version, please select GrxCAD.Interop.dll in inc-x64）.



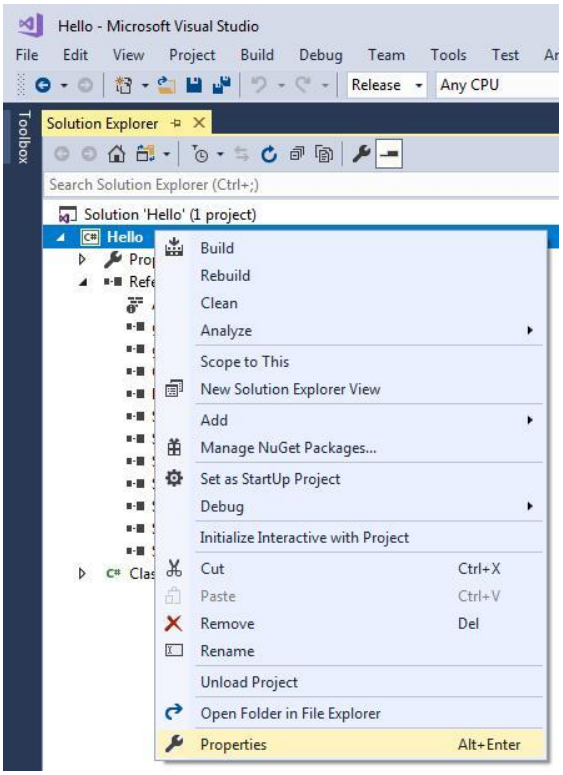


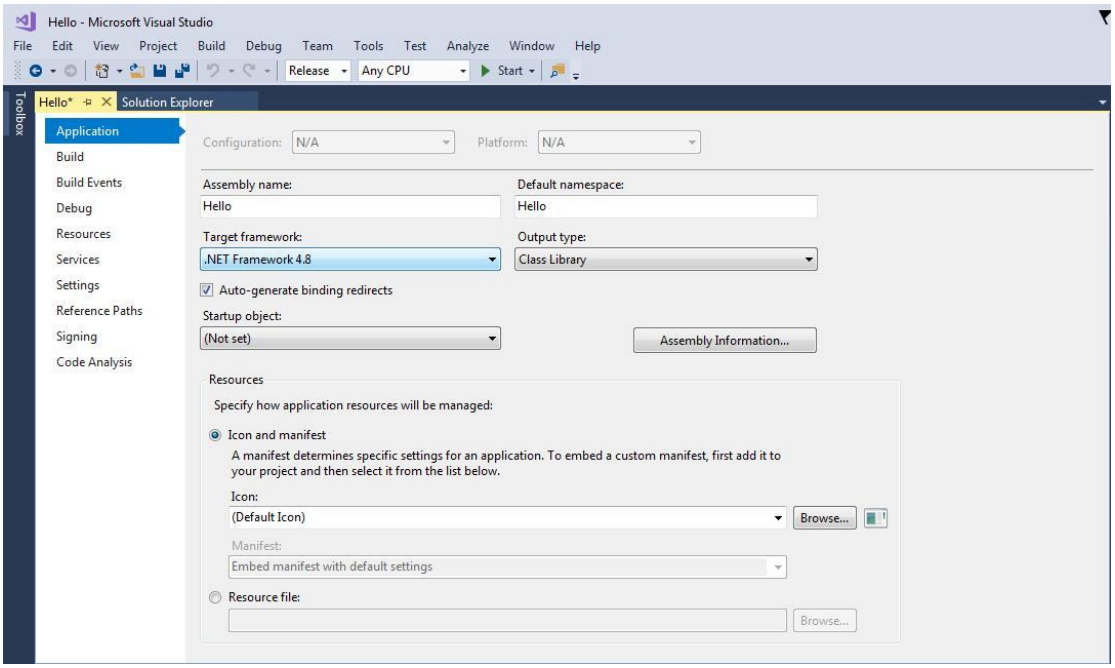
Remark：gmap.dll and gmdb.dll have to be added but file GrxCAD.Interop.dll is optional. After adding the files, right click each added reference file, click Property at the context menu, then the "Reference Properties" palette pops up, change the "Copy Local" property from True to False. The following image shows how to change "Copy Local" property for gmap.dll file. Please take it as reference to change gmdb.dll file's Copy Local property.



3.1.4. Change the Target Framework

In VS 2017, find the Hello project you have been created on the solution explorer. Right click to select Properties. After the new page opens, please change the "Target framework" to Framework 4.8 from the pull-down list in Application menu.





3.2. Compile Program

Click Visual Studio 2017 menu item [Build] » [Build solution], then you will get a “Hello.dll” file from D:\test\Hello\Hello\bin\Release directory.

Run GstarCAD and input "netload" at command line, there will a "Select .Net Assembly" dialog box pops out, select the "Hello.dll" file and click Open button to load it

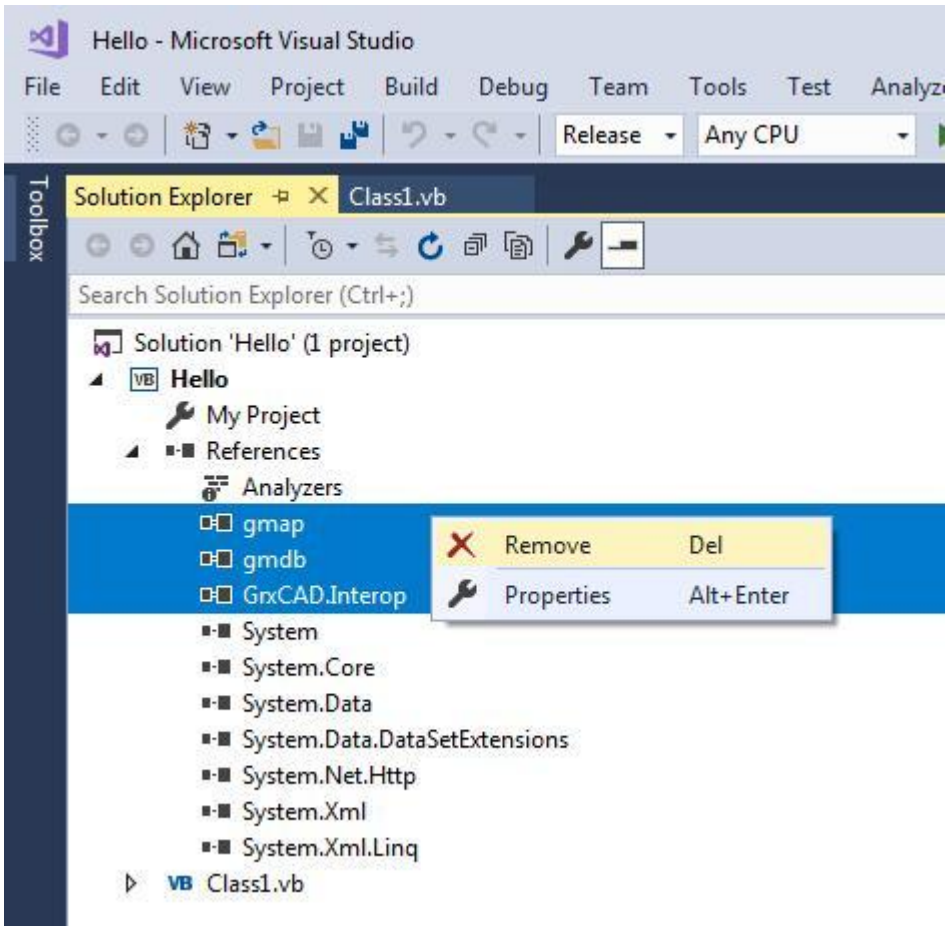
4. VS2017 VB.NET Upgrade Process

The following is a sample to create a "Hello" project, describe the creation process and suppose GstarCAD SDK installed in c:\grxsdk dictionary.

4.1. Change Reference Files

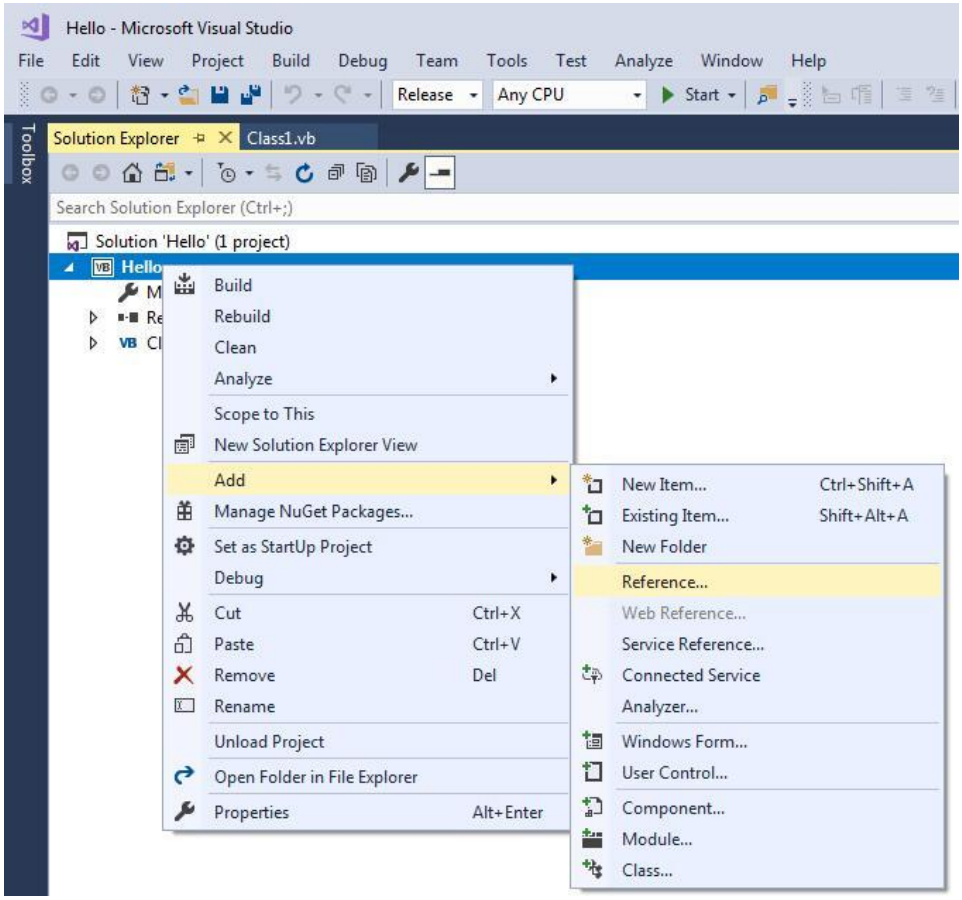
4.1.1. Remove the Reference

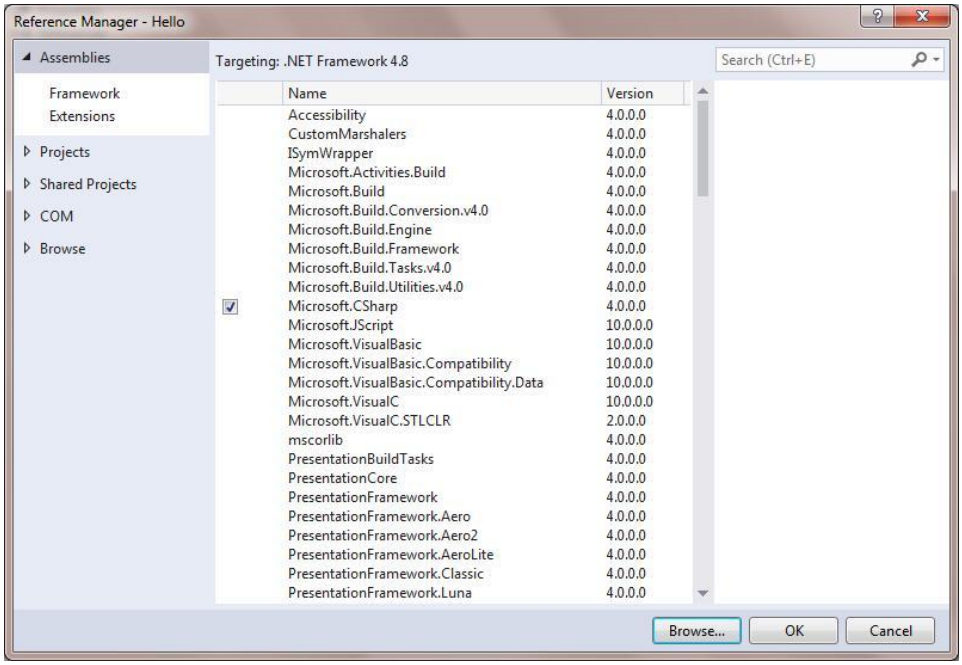
Right click on the reference files that referenced to the previous version of GstarCAD, click Remove.



4.1.2. Add Reference

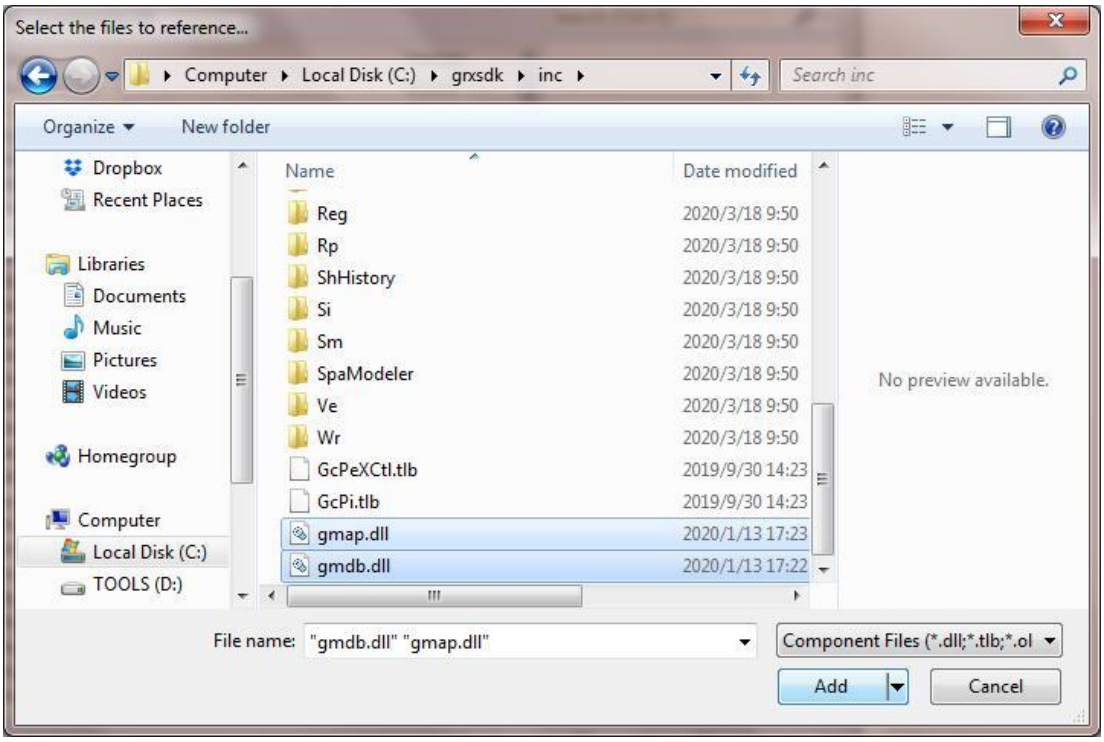
In VS 2017, find the Hello project you have been created on the solution explorer. Select the Hello project and right click to select Add Reference. After the "Reference Manager" dialog box pops out, click Browse button as shown below.

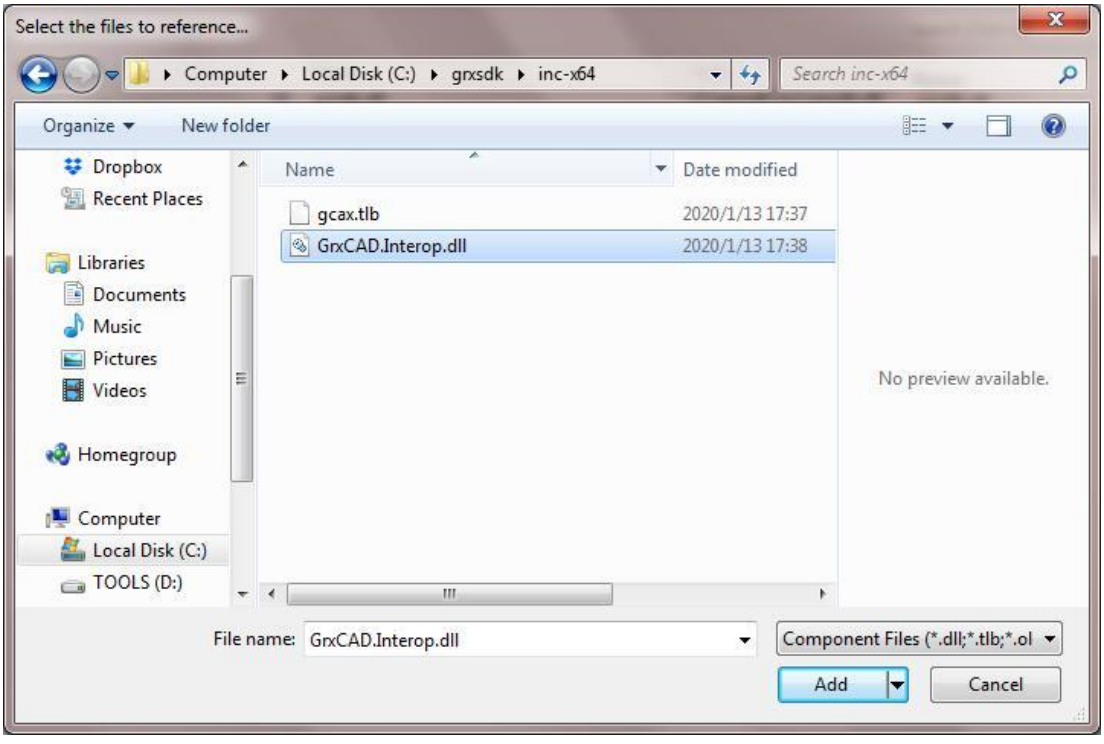

Hello - Microsoft Visual Studio
File Edit View Project Build Debug Team Tools Test Analyze Window Help
Release
Any CPU
Start
Toolbox
Solution Explorer
Class1.vb
Search Solution Explorer (Ctrl+;)
Solution 'Hello' (1 project)
Build
Rebuild
Clean
Analyze
Scope to This
New Solution Explorer View
Add
Manage NuGet Packages...
Set as StartUp Project
Debug
Cut Ctrl+X
Paste Ctrl+V
Remove Del
Rename
Unload Project
Open Folder in File Explorer
Properties Alt+Enter
New Item... Ctrl+Shift+A
Existing Item... Shift+Alt+A
New Folder
Reference...
Web Reference...
Service Reference...
Connected Service
Analyzer...
Windows Form...
User Control...
Component...
Module...
Class...


Reference Manager - Hello
Assemblies
Framework
Extensions
Projects
Shared Projects
COM
Browse
Targeting: .NET Framework 4.8
Search (Ctrl+E)
Name
Version
Accessibility 4.0.0.0
CustomMarshalers 4.0.0.0
ISymWrapper 4.0.0.0
Microsoft.Activities.Build 4.0.0.0
Microsoft.Build 4.0.0.0
Microsoft.Build.Conversion.v4.0 4.0.0.0
Microsoft.Build.Engine 4.0.0.0
Microsoft.Build.Framework 4.0.0.0
Microsoft.Build.Tasks.v4.0 4.0.0.0
Microsoft.Build.Utilities.v4.0 4.0.0.0
Microsoft.CSharp 4.0.0.0
Microsoft.JScript 10.0.0.0
Microsoft.VisualBasic 10.0.0.0
Microsoft.VisualBasic.Compatibility 10.0.0.0
Microsoft.VisualBasic.Compatibility.Data 10.0.0.0
Microsoft.VisualC 10.0.0.0
Microsoft.VisualC.STLCLR 2.0.0.0
mscorlib 4.0.0.0
PresentationBuildTasks 4.0.0.0
PresentationCore 4.0.0.0
PresentationFramework 4.0.0.0
PresentationFramework.Aero 4.0.0.0
PresentationFramework.Aero2 4.0.0.0
PresentationFramework.AeroLite 4.0.0.0
PresentationFramework.Classic 4.0.0.0
PresentationFramework.Luna 4.0.0.0
Browse...
OK
Cancel

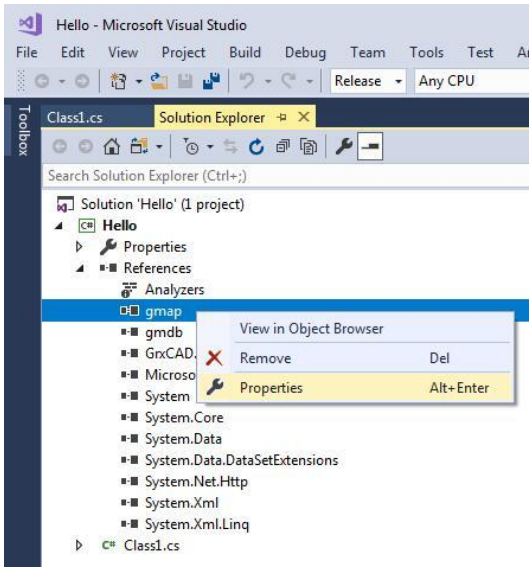
4.1.3. Choose and Add the Files

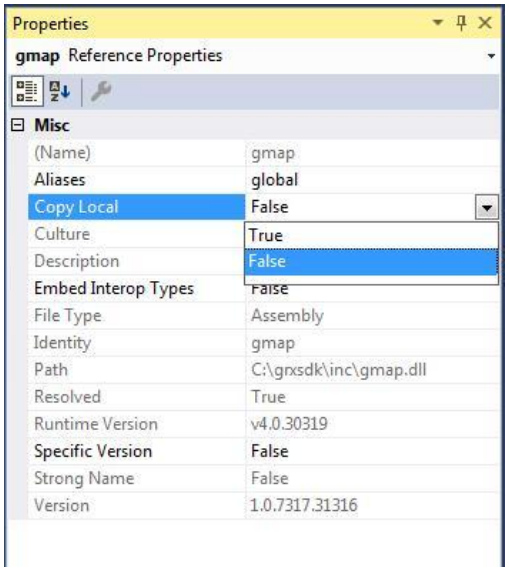
Find the GstarCADSDK installation path, for example, “C:\grxsdk\inc”. Select gmap.dll and gmdb.dll files in “c:\grxsdk\inc-x86” and add them by clicking Add button（If you use GstarCAD 64-bit version, please select GrxCAD.Interop.dll in inc-x64）





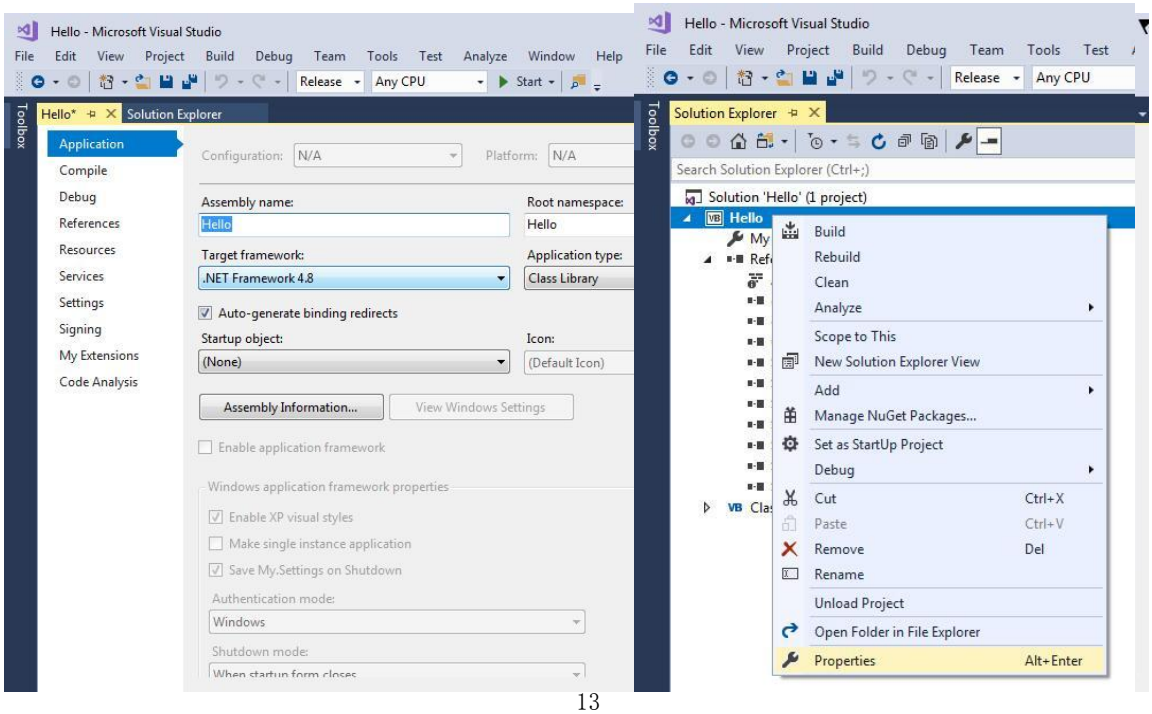
Remark：gmap.dll and gmdb.dll have to be added but file GrxCAD.Interop.dll is optional. After adding the files, right click each added reference file, click Property at the context menu, then the "Reference Properties" palette pops up, change the "Copy Local" property from True to False. The following image shows how to change "Copy Local" property for gmap.dll file. Please take it as reference to change gmdb.dll file's Copy Local property.





4.1.4. Change the Target Frame

In VS 2017, find the Hello project you have been created on the solution explorer. Right click to select Properties. After the new page opens, please change the "Target framework" to Framework 4.8 from the pull-down list in Application menu.



4.2. Compile Program

Click Visual Studio 2017 menu item [Build] » [Build solution], then you will get a “Hello.vb” file at the corresponding directory.

Run GstarCAD and input "netload" at command line, there will a "Select .Net Assembly" dialog box pops out, select the "Hello.vb" file and click Open button to load it

5. Copyright

Copyright reserved: Xi'an Gstarsoft Co.,Ltd

Usage License: Allows copying, quote any part of this document. Changes of any part of this document are forbidden without authorization. Please keep this statement when copying and quoting it, otherwise it will be held liable.

* AutoCAD® is a product of Autodesk® Company which is one of the CAD software solution providers. ARX is an abbreviated name of ObjectARX, which is AutoCAD® Runtime eXtension, and C++ SDK provided by Autodesk®.

.